

Marc Augustine Sojerido Amil

[LinkedIn](#) | +44 7426020205 | marc.amil@kcl.ac.uk | [Substack](#)

EDUCATION & AWARDS

King's College London

London, UK

PhD student specializing in generative AI in synthetic biology

2024 - 2027

- PhD Project: Generative models for *de novo* DNA design and functional insights in synthetic biology

Imperial College London

London, UK

MRes Neurotechnology (Bioengineering Department) Distinction

2022 - 2023

- Research Project: Living electrode-on-a-chip: developing a microphysiological model to study bioelectronic interfaces *in vitro*

Imperial College London

London, UK

BSc Biotechnology with Research Abroad, Upper Second Class Honours

2018 - 2022

- 2nd-year Dissertation: Deep Learning in Protein Structure Prediction and Design (82%)
- Final-year Dissertation: Closed-Loop System with DoE to Optimize Media for Cyanobacterial Growth (72%)

Grants: SynbiCITE Proof-of-Concept Grant (£50,000), iGEM Team Impact Grant (\$2500), The Francis Crick University Partnership Networking Fund (£4,000) — *Generative Biology: Unlocking the Future of Biosciences with Generative AI*, Lonza BioIncubate Program (CHF 10,000) — *Developed a pH feedback control loop for miniaturized cell culture systems bridging high-throughput biology and real-time control engineering*

Academic Awards: The President Undergraduate Scholarship, Faculty of Natural Sciences Make-A-Difference Runners Up 2022, Swiss European Mobility Programme Scholarship.

iGEM 2022 Awards: Top 10 Undergraduate Teams, Gold Medal, Nominations: Best Wiki, Best Food & Nutrition Project, Best Entrepreneurship.

SKILLS

Programming Languages: Python, R, HTML, CSS, MATLAB, bash

Software: LaTeX, GitHub, NumPy, Microsoft Office, COMSOL Multiphysics, Pandas, BioPython, Codex, Claude Code

Languages: English (native), Cebuano (native)

TECHNICAL EXPERIENCE

King's College London (Dr. Aleksej Zelezniak)

London, UK

PhD Researcher

April 2024 - Present

- Developed state-of-the-art (SOTA) large language models (LLMs) to predict gene expression in yeast by leveraging embeddings from pre-trained genomic language models focused on regulatory elements.
- Applied Deepmind's Alphagenome to identify and optimize DNA regulatory elements driving stable gene expression across multiple human cell types, achieving reduced sequence length with preserved function.

University College London (Dr. Duygu Dikicioglu)

London, UK

PhD Researcher

September 2023 - March 2024

- Developed CLAM, a Python-based computational pipeline combining *in silico* peptide cleavage simulation and antimicrobial peptide prediction to screen 1,011 *S. cerevisiae* genomes to identify novel antimicrobial peptides (AMP).
- Fine-tuned ESM2 for multi-label AMP activity prediction across 22 functional classes, achieving up to 99.23% accuracy for anti-MRSA and 98.92% for antibiofilm activity, outperforming existing classifiers.

International Genetically Engineered Machine (iGEM)

London, UK

Team member specializing in computational modelling

November 2021 - October 2022

- Developed a novel open-source software tool ([InFinity 1.0](#)) within 3 weeks that aims to increase protein affinity to a specific ligand in a high-throughput manner.
- Lead web designer for the [Wiki](#) which implemented HTML/CSS/Javascript. This was nominated Best Wiki out of 200+ candidates.
- Assisted in Grant writeup and fundraising, for which we raised ~£80,000 in 5 months. Additionally, educated the public about [Biotechnology through the EraCoBiotech competition](#), in which we achieved 3rd place.

Imperial College London (Prof. Rylie Green's Lab)

London, UK

MRes Student

October 2022 - Present

- Designed a microfluidic device to study neural interfaces *in vitro* using iPSC-derived neurons.
- Utilized advanced imaging techniques, including confocal microscopy, to visualize neuronal differentiation and neuronal innervation; also fabricated hydrogel scaffolds for stem cell development.

Nutrix

Poland

Biosensor Engineer

September 2021- March 2024

- Analyzed and collected 1500+ time-series glucose datasets and supported ML model development for glucose prediction using Python.
- Utilized lithography and microfabrication techniques to fabricate gold-based components for cortisol sensor development ([cortiSense](#)).

TEAMWORK & COMMUNICATION EXPERIENCE

King's College London x Francis Crick Institute

London, UK

Francis Crick University Partnership Networking Fund

January 2026 - Present

- Awarded a £4,000 competitive grant to establish a London AI × DNA research community, organising a 60+ participant AlphaGenome-focused hackathon and coordinating researchers from KCL, UCL, Imperial, and the Francis Crick Institute.

International Genetically Engineered Machine (iGEM)

London, UK

Team member specializing in computational Modelling and outreach November 2021 - October 2022

- Coordinated stakeholder engagement and successfully pitched to investors, raising £50,000+ in 3 months.
- Conducted field visits to UK farmers and food producers to assess the challenges in wheat production.

Imperial Enterprise Lab

London, UK

Global Challenge Lab Competition

July 2021 - August 2021

- Pitched an innovative product that combined IoT, AI and app development to tackle the technological issue of dengue. We were awarded 2nd place out of 700+ participants, additionally winning 1st place in the Communicable Diseases Category.

HACKATHON PROJECTS

Genomic Intelligence Builder Challenge — 1st Place

London, UK

[GI × Proto: Turning a Genomic Model into a Sequence Designer](#)

July 2026

- Won 1st place by building a closed-loop AI system combining Genomic Intelligence with Proto to optimize DNA sequences for expression and cell-type specificity.
- Demonstrated the approach across 50 human genes and 1,500 enhancers, improving predicted expression in 49/50 genes and steering enhancer specificity between HepG2 and K562 cell types

PUBLICATION & BOOK

Amil, M., Ventimiglia, L.N. and Zelezniak, A. (2026). Machine-Guided Approaches for Synthetic Biology Part Design. In Machine Learning and Big Data-enabled Biotechnology, H.S. Alper (Ed.). <https://doi.org/10.1002/9783527850532.ch3>

Bamezai, S., Maresca di Serracapriola, G., Morris, F., Hildebrandt, R., Amil, M. A. S., Sporadic iGEM Team, ... & Ledesma-Amaro, R. (2023). Protein engineering in the computational age: An open source framework for exploring mutational landscapes in silico. *Engineering Biology*.